

D862

Astrophysics with Lab

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word _____	The one fact that decides _____	Answer _____	[] sure [] unsure [] guess
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Items

1.

A virtual lab, part of a course that uses a free online telescope archive, gives a star's spectrum a peak at 500 nm. The star is in a cluster of about 300 stars. Using Wien's law ($\lambda_{\text{max}} \cdot T = 2.9 \times 10^{-3} \text{ m} \cdot \text{K}$), what is its surface temperature?

- A. $1.45 \times 10^{-9} \text{ K}$
- B. $5.8 \times 10^{-6} \text{ K}$
- C. 5,800 K
- D. 580 K

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2. [TRANSFER — reversed stem / different variable]

A star has a surface temperature of 29,000 K. At about what wavelength does its emission peak?

- A. 100 nm (ultraviolet)
- B. 1,000 nm
- C. 450 nm, since hot stars look blue.
- D. About 84 m

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3.

An exoplanet-survey team working from a remote desert station finds a star with the same radius as the Sun but twice the Sun's surface temperature. The station has a new solar array. How does the star's luminosity compare with the Sun's?

- A. 4 times greater
- B. 16 times greater
- C. 2 times greater
- D. 1/16 as bright

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4. [TRANSFER — reversed stem / different variable]

Two stars have the same surface temperature, but Star B has 3 times the radius of Star A. How many times more luminous is Star B?

- A.** 9 times
- B.** 27 times
- C.** 3 times
- D.** 81 times

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5.

An astronomy class using data from a public sky survey measures a galaxy receding at 7,000 km/s. The galaxy is a barred spiral with bright blue arms. Using $H_0 = 70$ km/s per Mpc, how far away is it?

- A.** 7,000 Mpc
- B.** 100 Mpc
- C.** 100 light-years
- D.** 490,000 Mpc

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6.

A radio-astronomy lab, using a dish array funded by a university consortium, measures hydrogen gas orbiting a spiral galaxy. Orbital speeds stay roughly flat far beyond the visible disk instead of falling off. The galaxy's bright core contains an old star cluster. What does this most strongly suggest?

- A.** Dark energy is pulling the outer gas outward.
- B.** Most of the mass is in the bright core.
- C.** The galaxy must follow a planet-like decline, so the data are wrong.
- D.** An extended halo of unseen (dark) matter adds mass far from the center.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7.

A supernova survey team, whose telescope time was awarded through a competitive review, finds that distant Type Ia supernovae look dimmer than expected for a universe whose expansion is slowing. The team observed 42 supernovae. What do the results indicate?

- A. Type Ia supernovae can never be used to measure distance.
- B. The expansion of the universe is accelerating, attributed to dark energy.
- C. The supernovae are closer than expected, so expansion is slowing.
- D. Dark matter is absorbing their light.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8.

A planetarium show produced by a science museum claims that if the Sun were replaced by a black hole of the same mass, Earth would be sucked in. The show runs 40 minutes. Which statement is correct?

- A. Earth would spiral into the black hole.
- B. Earth would fly off, because black holes have less gravity.
- C. The orbit would shrink because the black hole is much smaller than the Sun.
- D. Earth's orbit would stay the same, because gravity at Earth's distance depends only on the mass.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9. [TRANSFER — reversed stem / different variable]

A black hole has an event-horizon radius of 90 km. Using about 3 km per solar mass, what is its mass?

- A. About 3 solar masses
- B. About 270 solar masses
- C. About 9.5 solar masses
- D. About 30 solar masses

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10.

A deep-field image from a space telescope, processed by a volunteer team, shows a distant galaxy stretched into arcs around a massive foreground galaxy cluster. The image took 12 days of exposure. What causes the arcs?

- A. The cluster's mass curves spacetime and bends the background galaxy's light (gravitational lensing).
- B. Only black holes can bend light, never clusters.
- C. Light refracts through the cluster's gas like a glass lens.
- D. The background galaxy is being torn apart by the cluster.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11.

A student lab team analyzing public data from a planet-hunting satellite sees a star's brightness dip by 1% every 3.5 days. The star's radius is about the Sun's. The satellite launched in 2018. What is the planet's radius as a fraction of its star's?

- A. 0.0001
- B. 3.5
- C. 0.1
- D. 0.01

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12. [TRANSFER — reversed stem / different variable]

A planet whose radius is one-twentieth of its star's radius transits the star. How much does the star's brightness dip?

- A. 5%
- B. It brightens by 0.25%.
- C. 0.25%
- D. 0.0025%

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer C Concept 1

A virtual lab, ~~part of a course that uses a free online telescope archive~~, gives a star's spectrum a peak at 500 nm. ~~The star is in a cluster of about 300 stars.~~ Using Wien's law ($\lambda_{\text{max}} \cdot T = 2.9 \times 10^{-3} \text{ m} \cdot \text{K}$), what is its surface temperature?

Target word: temperature from peak wavelength **Decides:** $500 \text{ nm} = 5.0 \times 10^{-7} \text{ m}$.

A	Inverted operation	Multiplies the constant by the wavelength.
B	Dropped element	Drops the conversion from nanometers to meters.
C	Correct	$2.9 \times 10^{-3} \div 5.0 \times 10^{-7} = 5,800 \text{ K}$.
D	Wrong stem number	Misplaces a power of ten in the conversion.

2. Answer A Concept 1 • transfer

A star has a surface temperature of 29,000 K. At about what wavelength does its emission peak?

Target word: peak wavelength from temperature **Decides:** $\lambda = 2.9 \times 10^{-3} \div 29,000$.

A	Correct	$2.9 \times 10^{-3} \div 2.9 \times 10^4 = 1.0 \times 10^{-7} \text{ m} = 100 \text{ nm}$.
B	Wrong stem number	Off by a power of ten.
C	Confusion set	Looking blue does not mean the peak is in visible blue; this star peaks in the UV.
D	Inverted operation	Multiplies the constant by the temperature.

3. Answer B Concept 2

An exoplanet-survey team ~~working from a remote desert station~~ finds a star with the same radius as the Sun but twice the Sun's surface temperature. ~~The station has a new solar array.~~ How does the star's luminosity compare with the Sun's?

Target word: luminosity ratio **Decides:** $L \propto T^4$ at equal radius.

A	Confusion set	Uses the R^2 exponent for temperature.
B	Correct	$2^4 = 16$.
C	Dropped element	Drops the fourth power.
D	Inverted operation	Hotter stars of the same size are brighter, not dimmer.

4. Answer A Concept 2 • transfer

Two stars have the same surface temperature, but Star B has 3 times the radius of Star A. How many times more luminous is Star B?

Target word: luminosity from radius **Decides:** $L \propto R^2$ at equal temperature.

A	Correct	$3^2 = 9$.
B	Adjacent category	Cubes the radius as if luminosity scaled with volume.
C	Dropped element	Drops the square.
D	Confusion set	Uses the T^4 exponent for radius.

5. Answer B Concept 3

An astronomy class ~~using data from a public sky survey~~ measures a galaxy receding at 7,000 km/s. ~~The galaxy is a barred spiral with bright blue arms.~~ Using $H_0 = 70$ km/s per Mpc, how far away is it?

Target word: distance from Hubble's law **Decides:** $d = v \div H_0$.

A	Dropped element	Drops H_0 .
B	Correct	$7,000 \div 70 = 100$ Mpc.
C	Confusion set	Right number, wrong unit; the result is in megaparsecs.
D	Inverted operation	Multiplies speed by H_0 .

6. Answer D Concept 4

A radio-astronomy lab, ~~using a dish array funded by a university consortium~~, measures hydrogen gas orbiting a spiral galaxy. Orbital speeds stay roughly flat far beyond the visible disk instead of falling off. ~~The galaxy's bright core contains an old star cluster.~~ What does this most strongly suggest?

Target word: meaning of flat rotation **Decides:** Speeds do not fall with distance.

A	Confusion set	Dark energy acts on cosmic scales; extra gravity here points to dark matter.
B	Inverted operation	If mass were concentrated in the core, outer speeds would fall.
C	Absolute language	Flat curves are seen in many galaxies; the data are not the problem.
D	Correct	Flat speeds require mass that keeps growing with radius.

7. Answer B Concept 5

A supernova survey team, ~~whose telescope time was awarded through a competitive review~~, finds that distant Type Ia supernovae look dimmer than expected for a universe whose expansion is slowing. ~~The team observed 42 supernovae.~~ What do the results indicate?

Target word: what dim standard candles mean **Decides:** Dimmer = farther than a slowing universe allows.

A	Absolute language	They are standard candles precisely because their peak brightness is predictable.
B	Correct	Farther-than-expected distances mean the expansion sped up.
C	Inverted operation	Dimmer standard candles are farther, not closer.
D	Confusion set	Dark matter does not absorb light; dark energy is the explanation.

8. Answer D Concept 6

A planetarium show ~~produced by a science museum~~ claims that if the Sun were replaced by a black hole of the same mass, Earth would be sucked in. ~~The show runs 40 minutes.~~ Which statement is correct?

Target word: gravity of a same-mass black hole **Decides:** Same mass, same distance.

A	Confusion set	Mixes up near-horizon behavior with gravity at 1 AU.
B	Inverted operation	Same mass means the same pull at Earth's distance.
C	Adjacent category	Size does not change gravity at a fixed distance from the center.
D	Correct	The black hole's extreme gravity matters only very close to it.

9. Answer D Concept 6 · transfer

A black hole has an event-horizon radius of 90 km. Using about 3 km per solar mass, what is its mass?

Target word: mass from horizon radius **Decides:** Radius scales directly with mass.

A	Dropped element	Drops the 90 km and reports the per-mass constant.
B	Inverted operation	Multiplies instead of dividing.
C	Confusion set	Takes a square root, as if radius scaled with the square of mass.
D	Correct	$90 \div 3 = 30$.

10. Answer A Concept 7

A deep-field image from a space telescope, ~~processed by a volunteer team~~, shows a distant galaxy stretched into arcs around a massive foreground galaxy cluster. ~~The image took 12 days of exposure.~~ What causes the arcs?

Target word: cause of the arcs **Decides:** Massive cluster in front of a distant galaxy.

A	Correct	General relativity predicts light bending around mass.
B	Absolute language	Any mass bends light; clusters are strong lenses.
C	Confusion set	The bending comes from gravity, not refraction.
D	Adjacent category	The galaxy is far behind the cluster; only its image is distorted.

11. Answer C Concept 8

A student lab team analyzing public data from a planet-hunting satellite sees a star's brightness dip by 1% every 3.5 days. The star's radius is about the Sun's. ~~The satellite launched in 2018.~~ What is the planet's radius as a fraction of its star's?

Target word: planet-to-star radius ratio **Decides:** $(R_p/R_s)^2 = 0.01$.

A	Inverted operation	Squares the depth instead of taking its root.
B	Camouflage	Uses the orbital period, which does not set planet size.
C	Correct	$\sqrt{0.01} = 0.1$.
D	Dropped element	Drops the square root.

12. Answer C Concept 8 · transfer

A planet whose radius is one-twentieth of its star's radius transits the star. How much does the star's brightness dip?

Target word: transit depth from radius ratio **Decides:** $(1/20)^2$.

A	Dropped element	Drops the square.
B	Inverted operation	A transit blocks light; the star dims.
C	Correct	$(1/20)^2 = 1/400 = 0.0025 = 0.25\%$.
D	Confusion set	Treats the fraction 0.0025 as a percent.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	7	
Wrong stem number	A figure copied from the prompt that is not the answer.	2	
Adjacent category	The right idea in the wrong bucket.	3	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	10	
Absolute language	Always, never, guarantee — where judgment applies.	3	
Confusion set	Every option is real; the test is exact role-matching.	10	
Camouflage	Answers a stem fact that was never the question.	1	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Wien's law	A star's peak emission wavelength is inversely proportional to its temperature: $\lambda_{\text{max}} \cdot T = 2.9 \times 10^{-3} \text{ m} \cdot \text{K}$, so hotter stars peak at shorter wavelengths.	Multiplying instead of dividing, or forgetting to convert nm to m.	A peak wavelength and a question about temperature (or the reverse).
2	Stefan-Boltzmann luminosity	Luminosity scales as radius squared times temperature to the fourth power: $L \propto R^2 T^4$.	Scaling luminosity linearly with temperature or radius.	Two stars compared by radius or temperature.
3	Hubble's law	Recession speed = $H_0 \times$ distance, so distance = speed $\div H_0$ ($H_0 \approx 70 \text{ km/s per Mpc}$).	Multiplying speed by H_0 , or dropping H_0 .	A galaxy's recession speed and H_0 .
4	Flat rotation curves imply dark matter	Orbital speeds that stay flat far beyond a galaxy's visible edge mean unseen mass extends outward in a dark-matter halo.	Expecting outer gas to slow like outer planets, or crediting dark energy.	Speeds that do not fall with distance from a galaxy's center.
5	Accelerating expansion and dark energy	Distant Type Ia supernovae (standard candles) look dimmer — farther — than a slowing universe predicts, so the expansion is accelerating, attributed to dark energy.	Mixing up dark energy with dark matter, or reading dimmer as closer.	Standard candles that look fainter than expected.
6	Black holes: gravity depends on mass and distance	Far from a black hole, its gravity equals that of any object of the same mass; its event-horizon radius scales with mass (about 3 km per solar mass).	Thinking black holes "suck in" everything regardless of distance.	A black hole swapped in for a star of the same mass.
7	Gravitational lensing	Mass curves spacetime, so light from background objects bends around massive foreground objects, forming arcs or multiple images.	Explaining the arcs as refraction through gas or physical tearing.	Arcs of a distant galaxy around a massive cluster.
8	Transit depth	The fractional dip in starlight during a transit equals (planet radius $\div$ star radius) ² .	Forgetting to square (or take the square root).	A percent dip in brightness and a question about planet size.

Course scope (public description): Explore the vast and fascinating realm of astrophysics with a hands-on approach to learning about celestial objects, the formation of the universe, and the theories that explain their behavior. Through interactive virtual labs and engaging multimedia resources, you will delve into the properties of stars, planets, galaxies, black holes, and the fundamental forces that govern their motion. Analyze cutting-edge theories of general relativity, dark matter, and dark energy while developing critical analytical skills to interpret astronomical data. This course offers a comprehensive understanding of the universe and prepares you to apply astrophysical concepts in real-world contexts.